ALERT meeting

Pull request status

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IJCLab

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Pull request status

- Incoming pull requests
  1. Add routine to clean bad AHDC hits and Use ATOF hits in the Kalman Filter
     - It is more difficult than expected
     - A lot of divergences of `coatjava/development`, especially with ALERTEngine and Track

Conversation 0 | Commits 49 | Checks 22 | Files changed 13

ftouchte commented on Feb 20 • edited — Member

Features:

- Make the Kalman Filter reusable
- Add a routine to clean bad AHDC hits
- Use ATOF wedge in the Kalman Filter
- Extends `AHDC::kftrack`
- Convenient use of the PID in RungeKutta.java

Code consistency:

- Create an interface for the hits used by the Kalman Filter : `KFHit`
  - AHDC hits are stored in the usual `Hit`
  - beamline, ATOF wedges/bars are stored in `RadialKFHit`
  - the method `KFitter.correct()` no longer needs to explicitly check if we works with an AHDC hits or not: the polymorphism do the job for use
- Remove redundancy : `KFitter` does not use another `Stepper` in its constructor (the initial `Stepper` and the initial state are linked)

This is draft...

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  2. Layer and wire alignment implementation
     - It is easier, but I need to validate the structure of the tables
     - For the wire alignment, the angles are relative to the layers

```
# geometry/alert/ahdc/layer_alignment
# sector, layer, angle_start, angle_end
 1   11   0.875600   0.814700
 1   21   0.965500   0.760600
 1   22   0.553000   0.373300
 1   31   0.738500   0.768000
 1   32   0.392600   0.443200
 1   41   0.479700   0.481400
 1   42   0.252700   0.327400
 1   51   0.467300   0.219300
```

```
# geometry/alert/ahdc/wire_alignment
# sector, layer, wire, angle
 1   11    1   0.000000
 1   11    2   0.000000
 1   11    3   0.000000
 1   11    4   0.000000
 1   11    5   0.000000
 1   11    6   0.000000
 1   11    7   0.000000
 1   11    8   0.000000
 1   11    9   0.000000
 1   11   10   0.000000
 1   11   11   0.000000
 1   11   12   0.000000
 1   11   13   0.000000
 ...
 1   51   95   0.000000
 1   51   96   0.000000
 1   51   97   0.000000
 1   51   98   0.000000
 1   51   99   0.000000
```

```
+
+            // !!!  Alignment correction
+            int row = AlertDCWireIdentifier.layer2number((superlayerId+1)*10 + (layerId+1));
+            double start_angle_correction = Math.toRadians(cp.getDouble("/geometry/alert/ahdc/layer_alignment/start_angle", row));
+            double end_angle_correction = Math.toRadians(cp.getDouble("/geometry/alert/ahdc/layer_alignment/end_angle", row));
```